

Supplementary Material: Prototype User Interface

A Secure and Auditable Framework for Legal Data Management: A Hybrid Approach Using Hyperledger Fabric and IPFS

This supplement collects the prototype interface screenshots referenced in the main text (Section “User Interface and Functional Showcase”). The audit and transparency views (ledger explorer and audit log) appear in the main text; the remaining screens are shown here.

The registration screen (Fig. 1a) collects a username, email address, password, and role selection before account creation. The dashboard (Fig. 1b) is the landing page, presenting the primary navigation cards for case work, documents, communication, and audit-oriented tasks. The case-management view (Fig. 1c) displays active matters as cards with case titles, client names, status labels, short descriptions, and a control for creating a new case.

The document views show the information exposed after a file is selected. The document-details dialog (Fig. 1d) lists the file name, associated case, owner, content identifier, transaction identifier, integrity hash, and available actions. The download dialog (Fig. 2a) shows the selected file, access status, wrapped-key reference, encrypted-file reference, password prompt, and separate controls for downloading the encrypted object or saving the decrypted copy.

Register

Full Name

Email Address

Password

Associate Lawyer

CREATE ACCOUNT

[Already have an account? Login Here](#)

(a) Registration

BlockChain

namare (associate) Blockchain Verified Sign Out

Dashboard

Overview of your recent legal activities and secure documents.

Cases Management

Track active legal cases, review deadlines, and assign associate tasks.

Secure Documents

Upload, verify, and digitally sign case files using our secure vault.

Communication

Encrypted messaging with your legal team and external stakeholders.

Audit Trails

View immutable blockchain logs for all document access and modifications.

(b) Dashboard

BlockChain

namare (associate) Blockchain Verified Logout

Case Management

Your Cases (3) Create New Case

Acquisition of Greenfield Organics by Global Food Inc. Active

Client: Global Food Inc.
Type: Corporate
Created: Nov 7, 2025

Handling all legal aspects of the merger, including due diligence, regulatory filings with the SEC, and contract drafting for asset transfer.

Estate of Eleanor Vance Probate Hearing Active

Client: Thomas Vance
Type: General
Created: Nov 7, 2025

Representation for the executor of the estate of Eleanor Vance. The case involves the validation of the will and distribution of assets.

Phoenix Corp vs. Apex Innovations Patent Infringement Active

Client: Phoenix Corporation
Type: Intellectual Property
Created: Nov 7, 2025

(c) Case management

Document Details

Access checked: RBAC On-chain ACL

FILE NAME: EvidenceDocument.pdf

OWNER: admin-firma

STATUS: AES-256 Encrypted On Blockchain

ON-CHAIN TXID: 23becd3988ef4e6d567c1f566b9915d1368bdcab803899546d8f3898b23e6c

PLAINTEXT SHA-256 (CLIENT-COMPUTED): e442731496c63927314388c68a2249c4dc28acdc3153387472238aeb288232

Hash computed locally before encryption. Used for integrity verification.

ENCRYPTED BLOB SHA-256: b2ea81e8bc3435279b6fec65d5b1f4e7fb18756abf4b749f81726d5fad373fb

KEY ENVELOPE (BYES P-256, FOR THIS USER): HelixOuPRLg1otTe... [Encrypted Key Envelope] ..-y75tLT12p44cgr48PQ==

Envelope stored server-side; only your client can unwrap it.

Client-side encryption

Decryption keys are recovered locally after access approval; the server stores only encrypted key tokens and ciphertext.

Verify Integrity Download

(d) Document details (TxID + integrity hash)

Figure 1: Prototype user interface: registration, dashboard, case management, and document details.

Download Document

Access checked: RBAC + On-chain ACL

Auto-unlocked (client-side).

Keys are recovered locally after authorization. They are never sent by the server.

Recovered document key (client-side)

EDoY5W7e.....f1QESg=

Reveal (10s)

Recovered locally after access approval. Not visible to server admins.

Recovered IV (client-side)

ua9IX/5.....Sr0L

Reveal (10s)

Unique initialization vector for this specific file version.

Advanced (Offline/Debug)

Only use this for offline decryption tools. Treat exposed keys as highly sensitive.

Copy Full KeyCopy Full IV

Decrypt local .enc file

Browse...No file selected.

Download Encrypted

Decrypt & Save

(a) Document download (ECIES-wrapped key unwrapped client-side)

Figure 2: Document download dialog.